

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 6492A

6 Credits: 4

Open Elective

Source-grounded

Introduction to IoT

- Provide basic concepts of Internet of Things.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 6492A is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 6492A.pdf from the uploaded official SITTTTR syllabus archive.

Course code and title: preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Microelectronics
Course code	6492A
Course title	Introduction to IoT
Semester	6 Credits: 4
Course category	Open Elective
Periods per week	4 (L:3 T:1 P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Provide basic concepts of Internet of Things.

Course outcomes

CO	Official outcome	Study level
CO1	12 Understanding Things (IoT) Illustrate the IoT applications using Embedded	See official syllabus
CO2	14 Applying Computing Board - Arduino Identify the Python programming constructs for	See official syllabus
CO3	12 Applying Illustrate the IoT applications using Embedded	See official syllabus
CO4	20 Understanding Computing Board - Raspberry PI	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 3
CO3	CO3 3
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Explain the fundamental concepts of Internet of Things (IoT)	5	As specified
Module 2	Arduino	4	As specified
Module 3	Identify the Python programming constructs for IoT	4	As specified
Module 4	Raspberry PI	3	As specified

MODULE 1

Explain the fundamental concepts of Internet of Things (IoT)

This module is presented from the official syllabus scope for Introduction to IoT. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the fundamental concepts of Internet of Things (IoT)
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

▼ 2 Understanding. (IoT). Interpret IoT hardware, protocols and

2 Understanding. (IoT). Interpret IoT hardware, protocols and is an official topic in Module 1 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding. (IoT). Interpret IoT hardware, protocols and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding. (iot). interpret iot hardware, protocols and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding. (IoT). Interpret IoT hardware, protocols and means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-2 Understanding. (IoT). Interpret IoT hardware, protocols and definition/meaning. steps, application. Technical terms English-.

▼ 3 Understanding. software using IoT Stack. Summarize different IoT enabling

3 Understanding. software using IoT Stack. Summarize different IoT enabling is an official topic in Module 1 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding. software using IoT Stack. Summarize different IoT enabling using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding. software using iot stack. summarize different iot enabling should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 3 Understanding. software using IoT Stack. Summarize different IoT enabling means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Understanding. software using IoT Stack. Summarize different IoT enabling definition/meaning. definition/meaning, steps, application. Technical terms English- Malayalam.

▼ 3 Understanding. technologies. Classify IoT based on the complexity to build

3 Understanding. technologies. Classify IoT based on the complexity to build is an official topic in Module 1 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding. technologies. Classify IoT based on the complexity to build using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding. technologies. classify iot based on the complexity to build should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 3 Understanding. technologies. Classify IoT based on the complexity to build means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Understanding. technologies. Classify IoT based on the complexity to build definition/meaning. , steps, application . Technical terms English- .

▼ 3 Understanding. and operate.

3 Understanding. and operate. is an official topic in Module 1 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding. and operate. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding. and operate. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding. and operate. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Understanding. and operate.

പ്രദേശം definition/meaning പ്രദേശം. പ്രദേശം പ്രദേശം പ്രദേശം, steps, application പ്രദേശം പ്രദേശം. Technical terms English- പ്രദേശം പ്രദേശം.

▼ Explain challenges in IoT 1 Understanding Introduction and Definition of IoT, Applications - Characteristics - Things in IoT - IoT Stack - IoT enabling Technologies - IoT challenges - IoT Levels Illustrate the IoT applications using Embedded Computing Board -

Explain challenges in IoT 1 Understanding Introduction and Definition of IoT, Applications - Characteristics - Things in IoT - IoT Stack - IoT enabling Technologies - IoT challenges - IoT Levels Illustrate the IoT applications using Embedded Computing Board - is an official topic in Module 1 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain challenges in IoT 1 Understanding Introduction and Definition of IoT, Applications - Characteristics - Things in IoT - IoT Stack - IoT enabling Technologies - IoT challenges - IoT Levels Illustrate the IoT applications using Embedded Computing Board - using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain challenges in iot 1 understanding introduction and definition of iot, applications – characteristics - things in iot - iot stack – iot enabling technologies – iot challenges – iot levels illustrate the iot applications using embedded computing board - should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain challenges in IoT 1 Understanding Introduction and Definition of IoT, Applications – Characteristics - Things in IoT - IoT Stack – IoT enabling Technologies – IoT challenges – IoT Levels Illustrate the IoT applications using Embedded Computing Board - means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain challenges in IoT 1 Understanding Introduction and Definition of IoT, Applications – Characteristics - Things in IoT - IoT Stack – IoT enabling Technologies – IoT challenges – IoT Levels Illustrate the IoT applications using Embedded Computing Board - definition/meaning. steps, application Technical terms English-

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Arduino

This module is presented from the official syllabus scope for Introduction to IoT. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Arduino
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Illustrate the working of sensors and actuators 2 Understanding Outline the features of embedded computing

Illustrate the working of sensors and actuators 2 Understanding Outline the features of embedded computing is an official topic in Module 2 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the working of sensors and actuators 2 Understanding Outline the features of embedded computing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the working of sensors and actuators 2 understanding outline the features of embedded computing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the working of sensors and actuators 2 Understanding Outline the features of embedded computing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Illustrate the working of sensors and actuators 2 Understanding Outline the features of embedded computing □□□□□□□□□□ □□□□ definition/meaning □□□□□□□□□□. □□□□□ □□□□□ □□□□□□, steps, application □□□□□ □□□□□□□□ □□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□.

▼ 3 Understanding boards - Arduino. Illustrate the interfacing of sensors and

3 Understanding boards - Arduino. Illustrate the interfacing of sensors and is an official topic in Module 2 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding boards - Arduino. Illustrate the interfacing of sensors and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding boards - arduino. illustrate the interfacing of sensors and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 3 Understanding boards - Arduino. Illustrate the interfacing of sensors and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 3 Understanding boards - Arduino. Illustrate the interfacing of sensors and definition/meaning. , steps, application . Technical terms English- .

▼ 5 Applying actuators with Arduino Illustrate the interaction with web page and

5 Applying actuators with Arduino Illustrate the interaction with web page and is an official topic in Module 2 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Applying actuators with Arduino Illustrate the interaction with web page and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 applying actuators with arduino illustrate the interaction with web page and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 5 Applying actuators with Arduino Illustrate the interaction with web page and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 5 Applying actuators with Arduino Illustrate the interaction with web page and definition/meaning . . . , steps, application Technical terms English-

▼ 4 Applying cloud Sensors and Actuators - Role of Sensors and Actuators in IoT - Working of Sensors and Actuators - Examples. Embedded Computing Boards - features and characteristics of Arduino/Node MCU, Interfacing with Arudino, Configure Arduino for IoT, Transferring data from Web and Cloud.

4 Applying cloud Sensors and Actuators - Role of Sensors and Actuators in IoT - Working of Sensors and Actuators - Examples. Embedded Computing Boards - features and characteristics of Arduino/Node MCU, Interfacing with Arudino, Configure Arduino for IoT, Transferring data from Web and Cloud. is an official topic in Module 2 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying cloud Sensors and Actuators - Role of Sensors and Actuators in IoT - Working of Sensors and Actuators - Examples. Embedded Computing Boards - features and characteristics of Arduino/Node MCU, Interfacing with Arudino, Configure Arduino for IoT, Transferring data from Web and Cloud. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying cloud sensors and actuators - role of sensors and actuators in iot - working of sensors and actuators - examples. embedded computing boards - features and characteristics of arduino/node mcu, interfacing with arduino, configure arduino for iot, transferring data from web and cloud. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying cloud Sensors and Actuators - Role of Sensors and Actuators in IoT - Working of Sensors and Actuators - Examples. Embedded Computing Boards - features and characteristics of Arduino/Node MCU, Interfacing with Arduino, Configure Arduino for IoT, Transferring data from Web and Cloud. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 4 Applying cloud Sensors and Actuators - Role of Sensors and Actuators in IoT - Working of Sensors and Actuators - Examples. Embedded Computing Boards - features and characteristics of Arduino/Node MCU, Interfacing with Arduino, Configure Arduino for IoT, Transferring data from Web and Cloud. definition/meaning , steps,

► **Official syllabus detail and terminology (expand in digital version)**

27/57

MODULE 3

Identify the Python programming constructs for IoT

This module is presented from the official syllabus scope for Introduction to IoT. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Identify the Python programming constructs for IoT
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 2 Understanding. Python Illustrate the working of control structures in

2 Understanding. Python Illustrate the working of control structures in is an official topic in Module 3 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding. Python Illustrate the working of control structures in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding. python illustrate the working of control structures in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 2 Understanding. Python Illustrate the working of control structures in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 2 Understanding. Python Illustrate the working of control structures in definition/meaning steps, application Technical terms English-

▼ 4 Applying. Python Develop python programs to solve simple

4 Applying. Python Develop python programs to solve simple is an official topic in Module 3 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying. Python Develop python programs to solve simple using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying. python develop python programs to solve simple should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying. Python Develop python programs to solve simple means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-4 Applying. Python Develop python programs to solve simple problems. definition/meaning, steps, application, Technical terms English- Malayalam.

▼ 4 Applying. applications

4 Applying. applications is an official topic in Module 3 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying. applications using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying. applications should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying. applications means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-4 Applying. applications
 definition/meaning . . . , steps, application Technical terms English-
 .

▼ Make use of Python packages for IoT 2 Applying. Python - variables and data types - executing a python program - control structures - python module - simple programs - python for IoT. Illustrate the IoT applications using Embedded Computing Board -

Make use of Python packages for IoT 2 Applying. Python - variables and data types - executing a python program - control structures - python module - simple programs - python for IoT. Illustrate the IoT applications using Embedded Computing Board - is an official topic in Module 3 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Make use of Python packages for IoT 2 Applying. Python - variables and data types - executing a python program - control structures - python module - simple programs - python for IoT. Illustrate the IoT applications using Embedded Computing Board - using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, make use of python packages for iot 2 applying. python - variables and data types - executing a python program - control structures - python module - simple programs - python for iot. illustrate the iot applications using embedded computing board - should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Make use of Python packages for IoT 2 Applying. Python - variables and data types - executing a python program - control structures - python module - simple programs - python for IoT. Illustrate the IoT applications using Embedded Computing Board - means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Make use of Python packages for IoT 2 Applying. Python - variables and data types - executing a python program - control structures - python module - simple programs - python for IoT. Illustrate the IoT applications using Embedded Computing Board -
 പരമ്പരാഗതമായി നിർവ്വചന/അർത്ഥം നിർവ്വചിക്കുക. നിർവ്വചനം നിർവ്വചിക്കുക
 നിർവ്വചനം, steps, application നിർവ്വചിക്കുക നിർവ്വചിക്കുക നിർവ്വചിക്കുക. Technical terms
 English-ൽ നിർവ്വചിക്കുക നിർവ്വചിക്കുക.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Raspberry PI

This module is presented from the official syllabus scope for Introduction to IoT. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Raspberry PI
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ 2 Understanding boards - Raspberry PI Illustrate the interfacing of basic sensors with

2 Understanding boards - Raspberry PI Illustrate the interfacing of basic sensors with is an official topic in Module 4 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding boards - Raspberry PI Illustrate the interfacing of basic sensors with using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding boards - raspberry pi illustrate the interfacing of basic sensors with should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What 2 Understanding boards - Raspberry PI Illustrate the interfacing of basic sensors with means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 2 Understanding boards - Raspberry PI Illustrate the interfacing of basic sensors with definition/meaning . . . , steps, application Technical terms English-

▼ 8 Understanding embedded computing board - Raspberry PI Case Study - the applications building with IoT - Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart

8 Understanding embedded computing board - Raspberry PI Case Study - the applications building with IoT - Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart is an official topic in Module 4 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 8 Understanding embedded computing board - Raspberry PI Case Study - the applications building with IoT - Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 8 understanding embedded computing board - raspberry pi case study - the applications building with iot - smart perishable tracking/smart transportation, smart healthcare, smart should be discussed with

attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 8 Understanding embedded computing board - Raspberry PI Case Study - the applications building with IoT - Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 8 Understanding embedded computing board - Raspberry PI Case Study - the applications building with IoT - Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart
 പര്യവേക്ഷണം definition/meaning പര്യവേക്ഷണം. പര്യവേക്ഷണം പര്യവേക്ഷണം
 പര്യവേക്ഷണം, steps, application പര്യവേക്ഷണം പര്യവേക്ഷണം പര്യവേക്ഷണം. Technical terms
 English- പര്യവേക്ഷണം പര്യവേക്ഷണം.

▼ **10 Understanding Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system etc Programming Raspberry Pi with python, Interfacing sensors and actuators with Raspberry PI. Case Study - Applications building with IoT- Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system. Text / Reference T/R Book Title/Author RMD Sundaram Shriram K Vasudevan, Abhishek S Nagarajan, Internet of T1 Things, Wiley Publications Vijay Madisetti, Arshdeep Bahga, Internet of Things: A Hands-On Approach, R1 Orient Blackswan Raj Kamal, Internet of Things : Architecture And Design Principles, McGraw R2 Hill Education R3 Gary Smart, Practical Python Programming for IoT, Packt Publishing R4 Marco Schwartz, Internet of Things with Arduino Cookbook, Packt Publishing Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/106/105/106105166/> 2 <https://www.raspberrypi.org/blog/getting-started-with-iot/> 3 <https://learn.adafruit.com/category/internet-of-things-iot> 4 <https://www.arduino.cc/en/IoT/HomePage> 5 <http://esp32.net/>**

10 Understanding Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system etc Programming Raspberry Pi with python, Interfacing sensors and actuators with Raspberry PI. Case Study - Applications building with IoT- Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system. Text / Reference T/R Book Title/Author RMD Sundaram Shriram K Vasudevan, Abhishek S Nagarajan, Internet of T1 Things, Wiley Publications Vijay Madisetti, Arshdeep Bahga, Internet of Things: A Hands-On Approach, R1 Orient Blackswan Raj Kamal, Internet of Things : Architecture And Design Principles, McGraw R2 Hill Education R3 Gary Smart, Practical Python Programming for IoT, Packt Publishing R4 Marco Schwartz, Internet of Things with Arduino Cookbook, Packt Publishing Online Resources Sl.No

Website Link 1 <https://nptel.ac.in/courses/106/105/106105166/> 2

<https://www.raspberrypi.org/blog/getting-started-with-iot/> 3

<https://learn.adafruit.com/category/internet-of-things-iot> 4

<https://www.arduino.cc/en/loT/HomePage> 5 <http://esp32.net/> is an official topic in Module 4 of Introduction to IoT. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 10 Understanding Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system etc Programming Raspberry Pi with python, Interfacing sensors and actuators with Raspberry Pi. Case Study - Applications building with IoT- Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system. Text / Reference T/R Book Title/Author RMD Sundaram Shriram K Vasudevan, Abhishek S Nagarajan, Internet of T1 Things, Wiley Publications Vijay Madiseti, Arshdeep Bahga, Internet of Things: A Hands-On Approach, R1 Orient Blackswan Raj Kamal, Internet of Things : Architecture And Design Principles, McGraw R2 Hill Education R3 Gary Smart, Practical Python Programming for IoT, Packt Publishing R4 Marco Schwartz, Internet of Things with Arduino Cookbook, Packt Publishing Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/106/105/106105166/> 2 <https://www.raspberrypi.org/blog/getting-started-with-iot/> 3 <https://learn.adafruit.com/category/internet-of-things-iot> 4 <https://www.arduino.cc/en/loT/HomePage> 5 <http://esp32.net/> using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 10 understanding lavatory maintenance, smart water through iot, smart warehouse monitoring, smart retail, smart driver assistance system etc programming raspberry pi with python, interfacing sensors and actuators with raspberry pi. case study - applications building with iot- smart perishable tracking/smart transportation, smart healthcare, smart lavatory maintenance, smart water through iot, smart warehouse monitoring, smart retail, smart driver assistance system. text / reference t/r book title/author rmd sundaram shriram k vasudevan, abhishek s nagarajan, internet of t1 things, wiley publications vijay madiseti, arshdeep bahga, internet of things: a hands-on approach, r1 orient blackswan raj kamal, internet of things : architecture and design principles, mcgraw r2 hill education r3 gary smart, practical python programming for iot, packt publishing r4 marco schwartz, internet of things with arduino cookbook, packt publishing online resources sl.no website link 1 <https://nptel.ac.in/courses/106/105/106105166/> 2 <https://www.raspberrypi.org/blog/getting-started-with-iot/> 3 <https://learn.adafruit.com/category/internet-of-things-iot> 4 <https://www.arduino.cc/en/iot/homepage> 5 <http://esp32.net/> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 10 Understanding Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system etc Programming Raspberry Pi with python, Interfacing sensors and actuators with Raspberry PI. Case Study - Applications building with IoT- Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system.

Aspect	What to prepare
	Text / Reference T/R Book Title/Author RMD Sundaram Shriram K Vasudevan, Abhishek S Nagarajan, Internet of T1 Things, Wiley Publications Vijay Madisetti, Arshdeep Bahga, Internet of Things: A Hands-On Approach, R1 Orient Blackswan Raj Kamal, Internet of Things : Architecture And Design Principles, McGraw R2 Hill Education R3 Gary Smart, Practical Python Programming for IoT, Packt Publishing R4 Marco Schwartz, Internet of Things with Arduino Cookbook, Packt Publishing Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106/105/106105166/ 2 https://www.raspberrypi.org/blog/getting-started-with-iot/ 3 https://learn.adafruit.com/category/internet-of-things-iot 4 https://www.arduino.cc/en/IoT/HomePage 5 http://esp32.net/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-10 Understanding Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system etc Programming Raspberry Pi with python, Interfacing sensors and actuators with Raspberry PI. Case Study - Applications building with IoT- Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system. Text / Reference T/R Book Title/Author RMD Sundaram Shriram K Vasudevan, Abhishek S Nagarajan, Internet of T1 Things, Wiley Publications Vijay Madisetti, Arshdeep Bahga, Internet of Things: A Hands-On

Approach, R1 Orient Blackswan Raj Kamal, Internet of Things : Architecture And Design Principles, McGraw R2 Hill Education R3 Gary Smart, Practical Python Programming for IoT, Packt Publishing R4 Marco Schwartz, Internet of Things with Arduino Cookbook, Packt Publishing Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/106/105/106105166/> 2 <https://www.raspberrypi.org/blog/getting-started-with-iot/> 3 <https://learn.adafruit.com/category/internet-of-things-iot> 4 <https://www.arduino.cc/en/IoT/HomePage> 5 <http://esp32.net/>
 ഐഐഐഐ definition/meaning
 ഐഐഐഐ.
 ഐഐഐഐ
 ഐഐഐഐ, steps, application
 ഐഐഐഐ
 ഐഐഐഐ. Technical terms English-
 ഐഐഐഐ.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE**Concept and formula bank****Answer structure**

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► **Define or explain 2 Understanding. (IoT). Interpret IoT hardware, protocols and. (3 marks)**

► **Define or explain 3 Understanding. software using IoT Stack. Summarize different IoT enabling. (3 marks)**

► **Define or explain 3 Understanding. technologies. Classify IoT based on the complexity to build. (3 marks)**

► **Define or explain 3 Understanding. and operate.. (3 marks)**

Module 2

► **Define or explain Illustrate the working of sensors and actuators 2 Understanding Outline the features of embedded computing. (3 marks)**

► **Define or explain 3 Understanding boards - Arduino. Illustrate the interfacing of sensors and. (3 marks)**

► **Define or explain 5 Applying actuators with Arduino Illustrate the interaction with web page and. (3 marks)**

► **Define or explain 4 Applying cloud Sensors and Actuators - Role of Sensors and Actuators in IoT - Working of Sensors and Actuators -**

Examples. Embedded Computing Boards - features and characteristics of Arduino/Node MCU, Interfacing with Arudino, Configure Arduino for IoT, Transferring data from Web and Cloud.. (3 marks)

Module 3

► **Define or explain 2 Understanding. Python Illustrate the working of control structures in. (3 marks)**

► **Define or explain 4 Applying. Python Develop python programs to solve simple. (3 marks)**

► **Define or explain 4 Applying. applications. (3 marks)**

► **Define or explain Make use of Python packages for IoT 2 Applying. Python - variables and data types - executing a python program - control structures - python module - simple programs - python for IoT. Illustrate the IoT applications using Embedded Computing Board -. (3 marks)**

Module 4

► **Define or explain 2 Understanding boards - Raspberry PI Illustrate the interfacing of basic sensors with. (3 marks)**

► **Define or explain 8 Understanding embedded computing board - Raspberry PI Case Study - the applications building with IoT - Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart. (3 marks)**

► **Define or explain 10 Understanding Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system etc Programming Raspberry Pi with python, Interfacing sensors and actuators with Raspberry PI. Case Study -**

Applications building with IoT- Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system. Text / Reference T/R Book Title/Author RMD Sundaram Shriram K Vasudevan, Abhishek S Nagarajan, Internet of T1 Things, Wiley Publications Vijay Madisetti, Arshdeep Bahga, Internet of Things: A Hands-On Approach, R1 Orient Blackswan Raj Kamal, Internet of Things : Architecture And Design Principles, McGraw R2 Hill Education R3 Gary Smart, Practical Python Programming for IoT, Packt Publishing R4 Marco Schwartz, Internet of Things with Arduino Cookbook, Packt Publishing Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/106/105/106105166/> 2 <https://www.raspberrypi.org/blog/getting-started-with-iot/> 3 <https://learn.adafruit.com/category/internet-of-things-iot> 4 <https://www.arduino.cc/en/IoT/HomePage> 5 <http://esp32.net/>. (3 marks)

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **2 Understanding. (IoT). Interpret IoT hardware, protocols and.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Illustrate the working of sensors and actuators 2 Understanding Outline the features of embedded computing.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **2 Understanding. Python Illustrate the working of control structures in.**

Module 4

1-mark definition: State the meaning of **2 Understanding boards - Raspberry PI Illustrate the interfacing of basic sensors with.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- <https://nptel.ac.in/courses/106/105/106105166/>
- <https://www.raspberrypi.org/blog/getting-started-with-iot/>
- <https://learn.adafruit.com/category/internet-of-things-iot>
<https://www.arduino.cc/en/IoT/HomePage> <http://esp32.net/>

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 6492A. Verify institutional updates against the current official curriculum.